Signal & Data Analysis in Neuroscience 2020

Systems

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1

Outline – Frequency Domain

- ☑ Introduction
- ☑ Fourier Transform
- ☑ Sampling Theory
- Systems
- Filters
- Spectral Analysis

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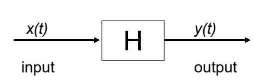
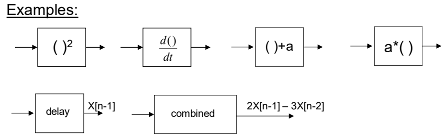
2

System definition

- Any manipulation of a signal can thought of as passing the signal through a **system** (H):



Examples:



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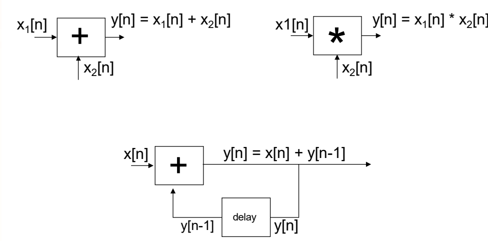
3

Common systems

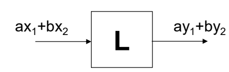


4

Linear Systems

- Consider a system **L**. Suppose the response of the system to inputs x_1 is y_1 and to input x_2 is y_2

- The system is linear iff:
  for every x_1,x_2,a,b $ax_1+bx_2 \rightarrow ay_1+by_2$

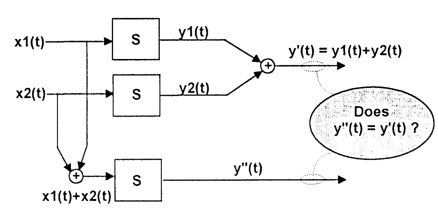


- Example of a *linear* system : y[n] = 2x[n]
- Example of a *nonlinear* system: y[n] = $(x[n])^2$

iff → if and only if

5

Checking Linearity

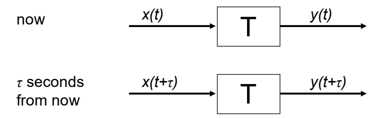


Taken from: http://www.ecse.rpi.edu/

6

Time Invariant (TI) systems

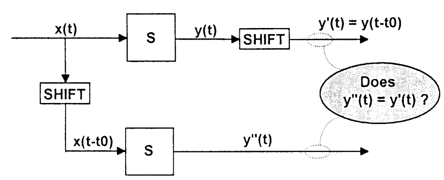
System is constant in time : Whether we apply an input to the system, **now** or **τ seconds from now**, the output will be identical, except a time delay of the τ seconds



Example of a *time invariant* system : y[n] = 2*x[n]
Example of a *time varying* system : y[n] = n*x[n]

7

Checking TI



Taken from: http://www.ecse.rpi.edu/

8

Linear time invariant (LTI) systems

- LTI → both **linear** and **time invariant**
- Can be represented by a series of *N* coefficients $\{h[n]\}_0^{N-1}$ such that:

  $$y[n] = h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + \ldots$$

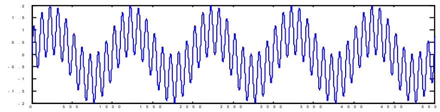
- It therefore performs the **convolution** of the signal with the systems' coefficients:

  $$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

9

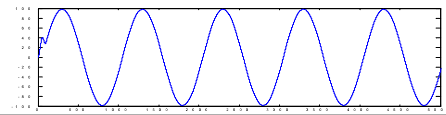
Rectangular convolution

$$signal = \sin(\frac{2\pi}{1000}t) + \sin(10 * \frac{2\pi}{1000}t)$$



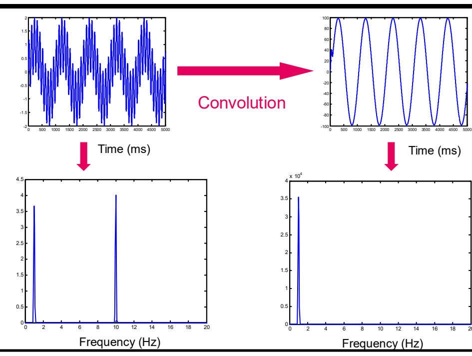
Convolution with a rectangular window

result = conv(ones(1,100), signal)



10

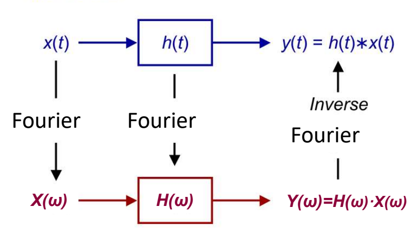
Fourier Transform



11

Time vs. Frequency domains

Time domain



Frequency domain

12

Shift theorem

$$\mathrm{DFT}_k(x[n-\Delta]) = \sum_{n=0}^{N-1} x[n-\Delta]e^{-j2\pi kn/N}$$

$$= \sum_{m=-\Delta}^{N-1-\Delta} x[m]e^{-j2\pi k(m+\Delta)/N} \quad (m = n-\Delta)$$

$$= \sum_{m=0}^{N-1} x[m]e^{-j2\pi km/N}e^{-j2\pi k\Delta/N}$$

$$= e^{-j2\pi k\Delta/N}\sum_{m=0}^{N-1} x[m]e^{-j2\pi km/N}$$

$$= e^{-j\omega_k\Delta}X[k]$$

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13

Convolution

$$\mathrm{DFT}_k(x*h) = \sum_{n=0}^{N-1} (x*h)_n\, e^{-j2\pi kn/N}$$

$$= \sum_{n=0}^{N-1}\sum_{m=0}^{N-1} x[m]h[n-m]e^{-j2\pi kn/N}$$

$$= \sum_{m=0}^{N-1} x[m]\sum_{n=0}^{N-1} h[n-m]e^{-j2\pi kn/N}$$

$$= \left(\sum_{m=0}^{N-1} x[m]e^{-j2\pi km/N}\right)H[k] \quad \text{(by the Shift Theorem)}$$

$= X[k]H[k]$

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14

Time and frequency domains

Time domain $\overset{\mathcal{F}}{\longleftrightarrow}$ Frequency Domain

Time domain	Frequency Domain
• x[n], x(t)	• X[k], X(ω)
• h(n)	• H(ω)
• convolution	• multiplication

IMPORTNAT

- All real systems are in the time domain!!!!
- No frequency domain systems exist.
- We just utilize frequency domain for our convenience

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15

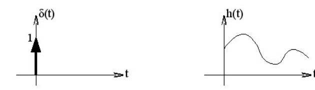
Impulse response

All LTI systems can be characterized entirely by a single function called the system's impulse response

$\delta(t) \longrightarrow \boxed{\mathcal{H}} \longrightarrow h(t)$

$$y[n] = x[n] * h[n]$$
$$x[n] = \delta[n]$$

$$y[n] = \delta[n] * h[n] = \sum_{k=-\infty}^{\infty} \delta[k] h[n-k] = h[n]$$



Richard Baraniuk, cnx.org

16

Transfer function

Equivalently, an LTI system can be characterized in the frequency domain by the transfer function H(ω):

- The relation between the input and output
  $$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$
- and also, the transform of the system's impulse response $H(\omega) = F(h[n])$
- Note: Throughout the slides we'll mix [ ] and ( ) and indexes but in all the following slides we'll deal with the discrete case i.e. DFT

17

FIR and IIR Systems

Systems can be divided into 2 groups, characterized by their impulse response function

FIR

Finite Impulse Response

$$\exists M . \forall h[n]\Big|_{|n|>M} = 0$$

$$H(\omega) = \frac{p(\omega)}{1}$$

IIR

Infinite Impulse Response

$$\{h[n]\}_{-\infty}^{\infty}$$

$$H(\omega) = \frac{p(\omega)}{q(\omega)}$$

18

LTI Systems – General Representation

Since h[n] in a LTI system are scalars, the system can be described by the **linear constant coefficients** difference equation:

$$y_{(\nu)} + \sum_{i=1}^{n} a_i \, y_{(\nu-i)} = \sum_{k=0}^{m} b_k \, x_{(\nu-k)}$$

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19

FIR Systems

$$y(n) = b_0 x(n) + b_1 x(n-1) + ... + b_P x(n-P)$$

- System order = number of coefficients (P)
- “Moving Average” like – convolution of signal with window

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20

FIR implementation

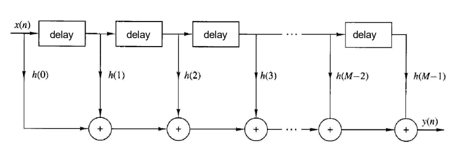

Figure 7.1 Direct-form realization of FIR system.

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21

FIR example

n	x(n)	y(n) Average of x(n) over last 5 samples
1	10	2
2	22	6.4
3	24	11.2
4	42	19.6
5	37	27
6	77	40.4
7	89	53.8
8	22	53.4
9	63	57.6
10	9	52

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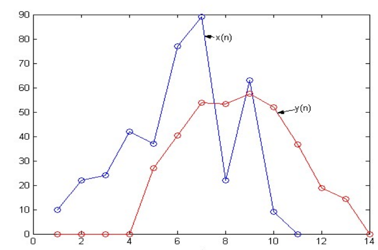
22

FIR example II



Observations:
1) There is a delay between input and output sequences
2) The output is more smooth as compared to the input
3) If the input goes to zero then the output follows suit after some time

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23

IIR systems

$$y(n) = \underbrace{b_0x(n)+b_1x(n-1)+\cdots+b_Px(n-P)}+\underbrace{a_1y(n-1)+a_2y(n-2)+\cdots+a_Qy(n-Q)}$$

$$y(n) = \sum_{i=0}^{P} b_i x(n-i) + \sum_{k=1}^{Q} a_k y(n-k)$$

- Involves a feedback system (memory)
- perturbations at the input could, depending upon the design, cause unstability (y→inf) and oscillations
- Require far fewer multiplications than FIR systems, to achieve similar response

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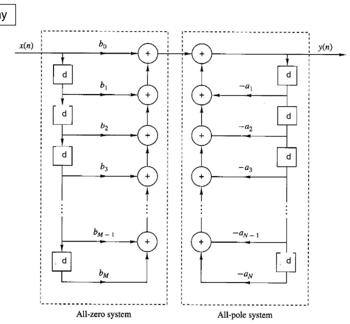
24

IIR implementation

d = delay



25

Recursive (IIR) system - example

y(n) = x(n) + y(n-1) → y(n)=x(n)+x(n-1)+x(n-2)+…+x(0)

Recursive definition → Infinite output response

What happens if x(0)=1 and x(n>1)=0 ???

26

IIR – exploding example

n	x(n)	y(n) = x(n) + 2y(n-1)
-6	0	0
-5	0	0
-4	0	0
-3	0	0
-2	0	0
-1	0	0
0	1	1
1	0	2
2	0	4
3	0	8
4	0	16
5	0	32
6	0	64
7	0	128

27

IIR – oscillating example

n	x(n)	y(n) = x(n) + -y(n-1)
-6	0	0
-5	0	0
-4	0	0
-3	0	0
-2	0	0
-1	0	0
0	1	1
1	0	-1
2	0	1
3	0	-1
4	0	1
5	0	-1
6	0	1
7	0	-1

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28

Example – Averaging 5 last inputs

- No feedback (FIR) ?
- With feedback (IIR) ?

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29

FIR averaging

n	x[n]	$y[n] = \frac{1}{5}x[n] + \frac{1}{5}x(n-1) + \frac{1}{5}x(n-2) + \frac{1}{5}x(n-3) + \frac{1}{5}x(n-4)$
-6	0	0
-5	0	0
-4	0	0
-3	0	0
-2	0	0
-1	0	0
0	1	1/5
1	0	1/5
2	0	1/5
3	0	1/5
4	0	1/5
5	0	0
6	0	0

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30

IIR averaging

n	x(n)	$y(n) = \frac{1}{5}x(n) + y(n-1) - \frac{1}{5}x(n-5)$
-6	0	0
-5	0	0
-4	0	0
-3	0	0
-2	0	0
-1	0	0
0	1	1/5
1	0	1/5
2	0	1/5
3	0	1/5
4	0	1/5
5	0	0
6	0	0

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31

System's Response

- Response both in **Amplitude** and **Phase**
- Amplitude response usually measured logarithmically – in dB units
- Every LTI system, $H(\omega) = A(\omega)e^{i\theta(\omega)}$,has the effect of delaying each sinusoidal component by : $\frac{\theta(\omega)}{\omega}$
- System's respose is presented either in log scale (Bode) or semi-log scale

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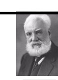
32

Decibel (dB)



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- The **decibel** (**dB**) is a measure of the ratio between two quantities:

$$X_{\mathrm{dB}} = 20\log_{10}\left(\frac{X}{X_0}\right)$$

- −3 dB = ½ power → difference of ±3 dB is roughly double/half power (actually 1.995).

Magnitude	$20\log_{10}$ G (dB)
100	40
10	20
1	0
0.1	-20
0.01	-40

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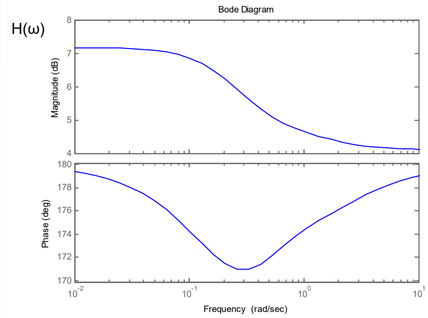
33

Bode Plot (fully logarithmic)



34

Frequency normalization

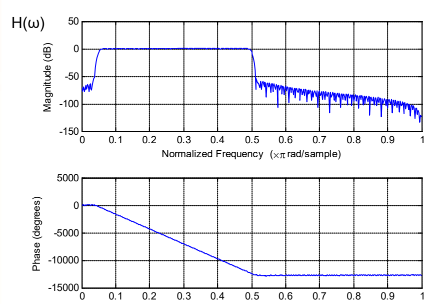
- During the analysis of sampled signals it is convenient in many cases to look at the frequencies over the range of 2π.

$$\omega = \frac{2\pi \cdot f}{f_s}$$

- As $2f_{max}<f_s \rightarrow \omega \le \pi$ (remember Nyquist)

35

Semi-Log plot



matlab: freqz

36